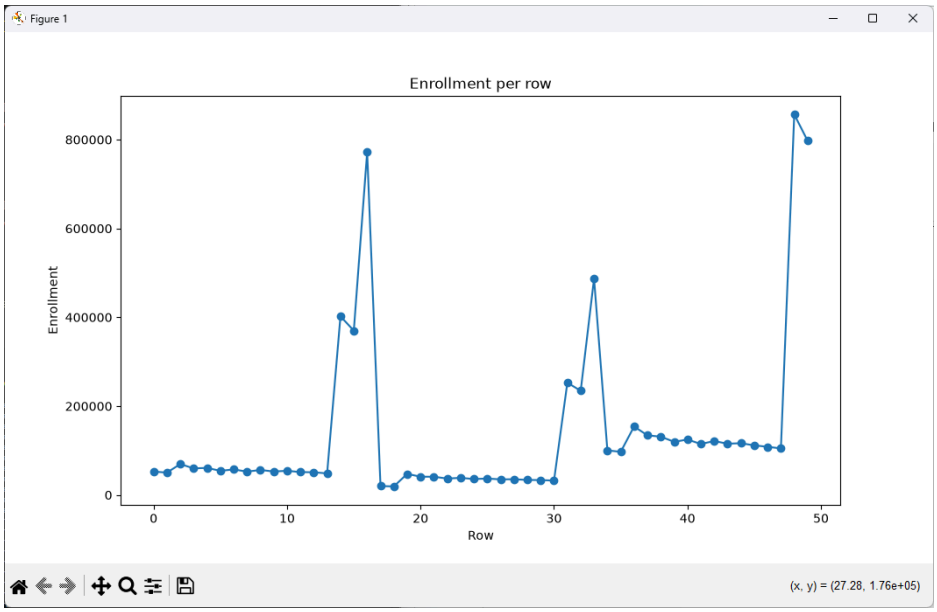
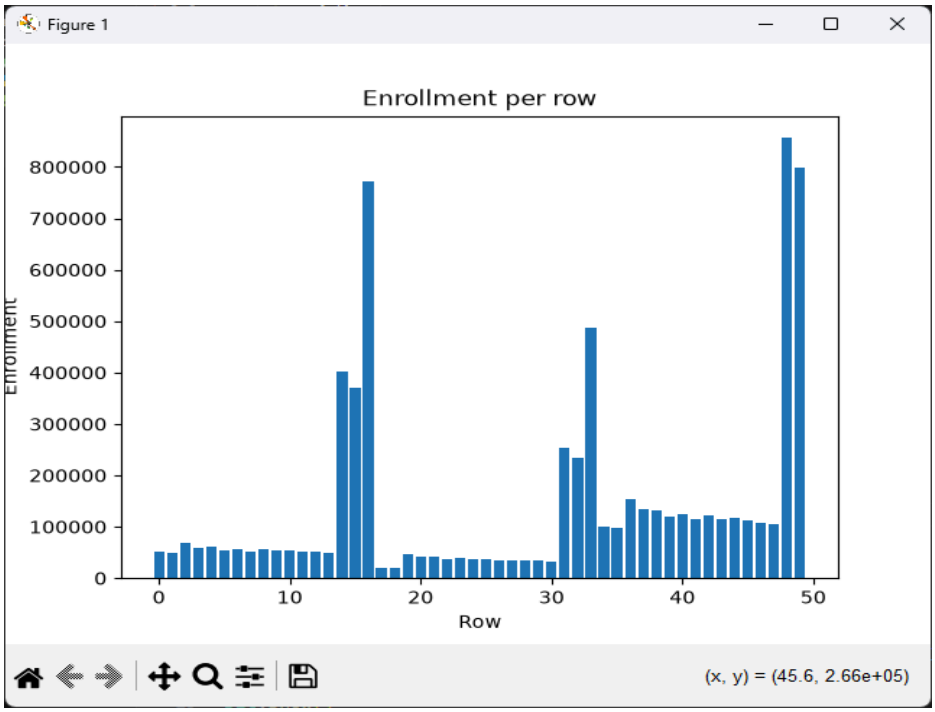
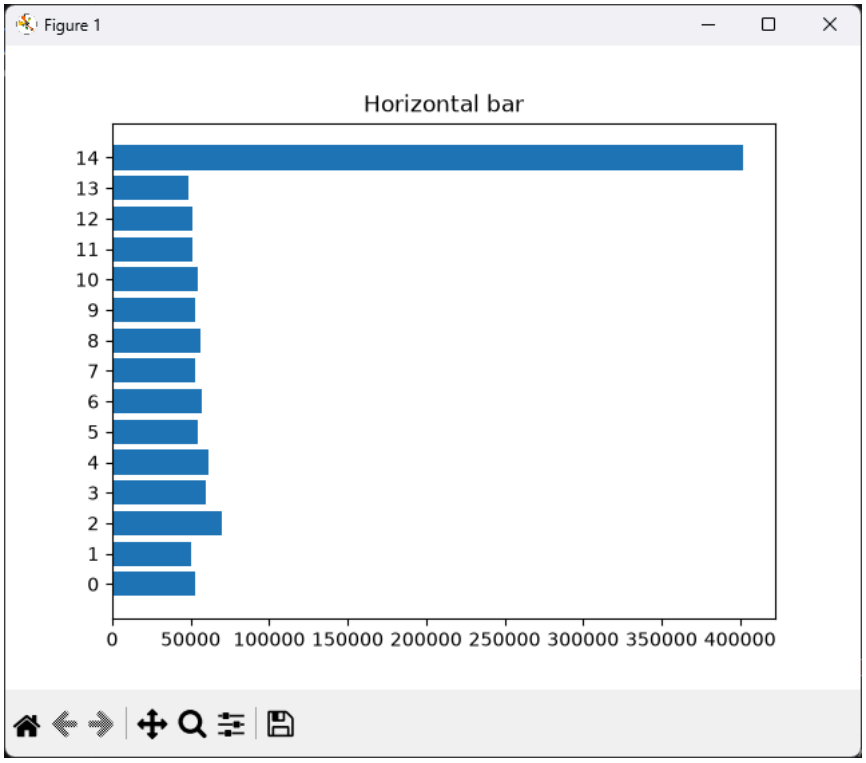
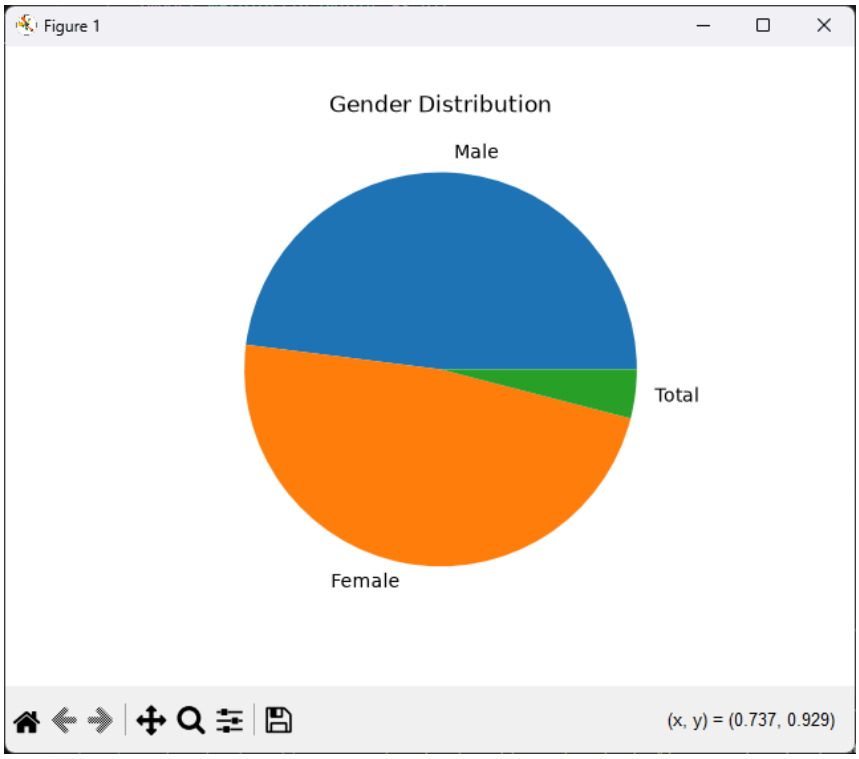
Lachica, Roniel Latint T. BSIT 4A - DA

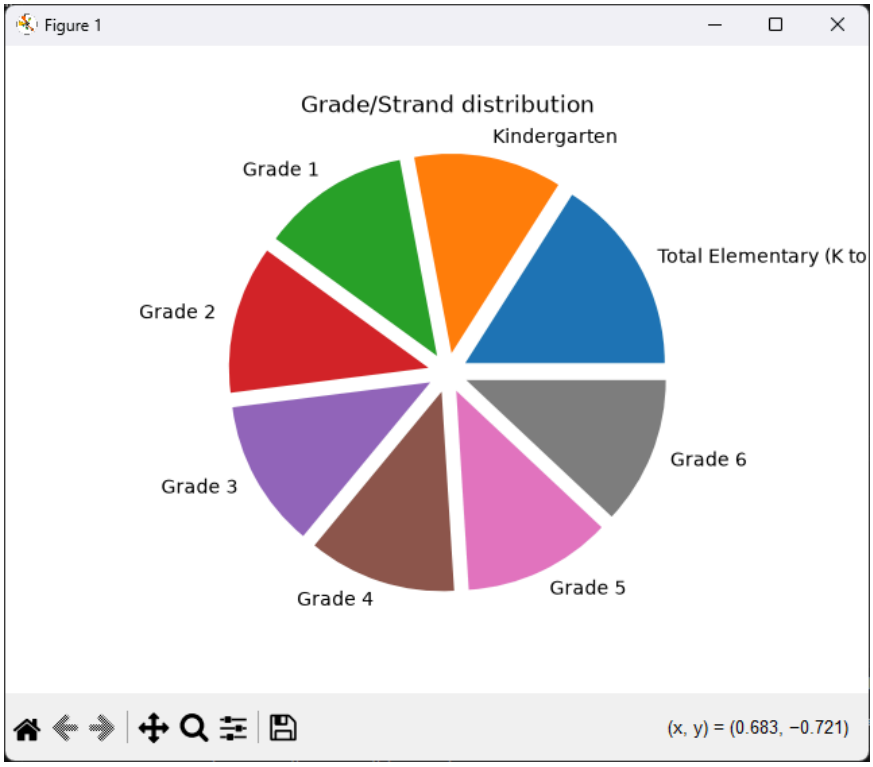
Laboratory Exercise # 1: Using Matplotlib

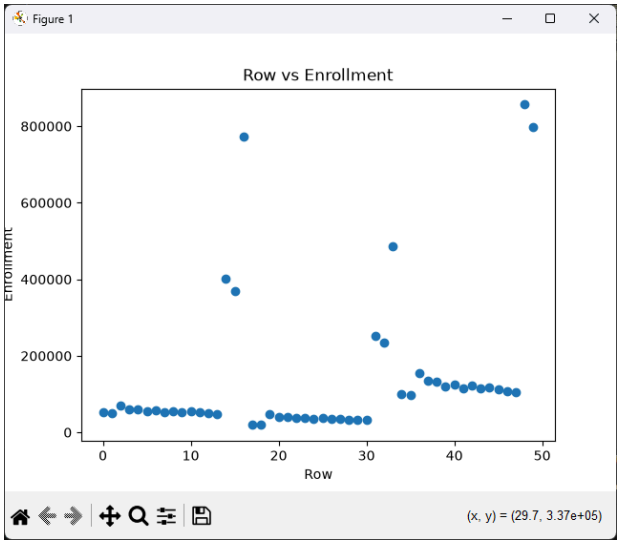


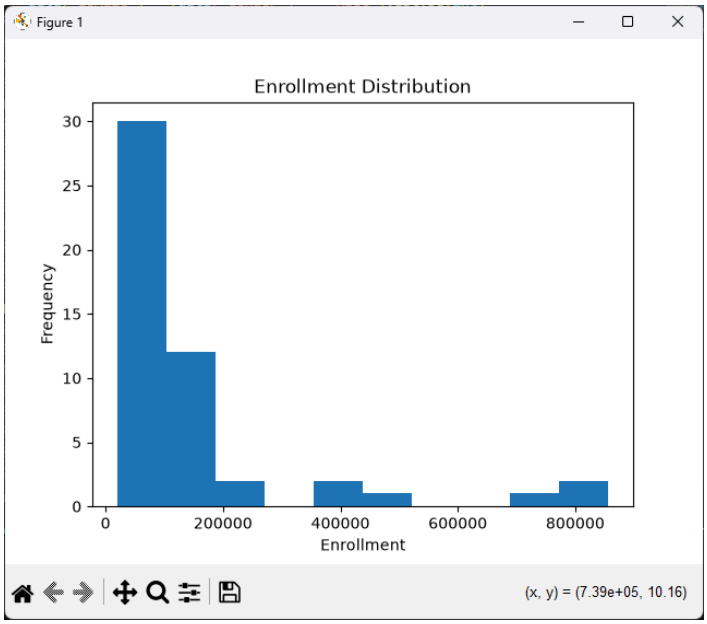


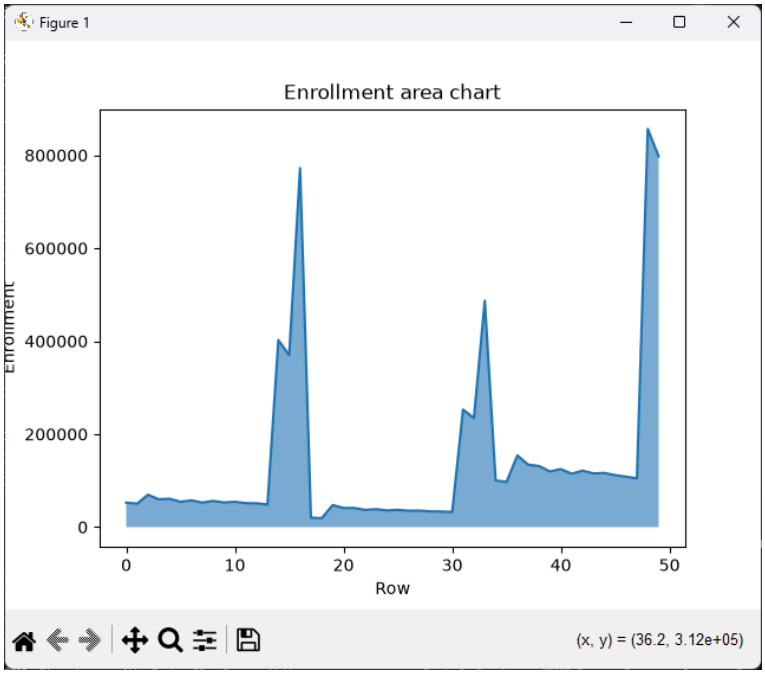

Figure 1
Horizontal bar
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
0
50000
100000
150000
200000
250000
300000
350000
400000

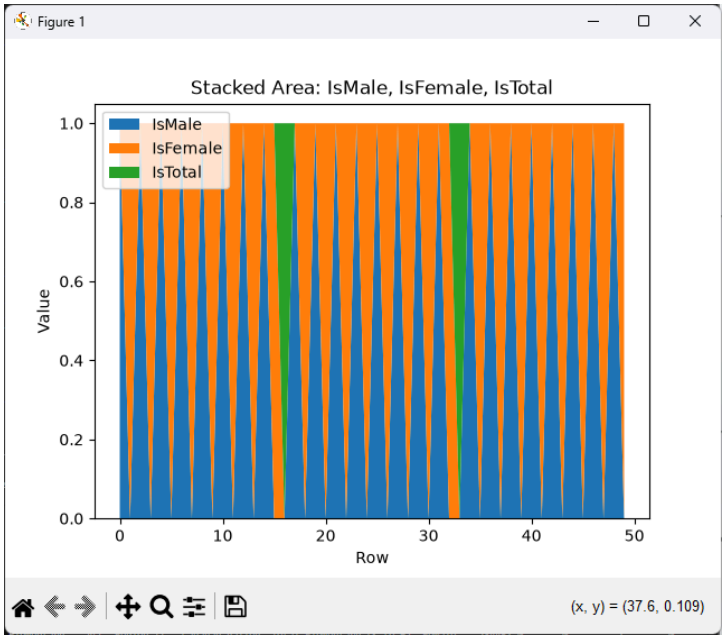

Figure 1
Gender Distribution
Male
Total
Female
(x, y) = (0.737, 0.929)


Figure 1
Grade/Strand distribution
Kindergarten
Grade 1
Total Elementary (K to
Grade 2
Grade 6
Grade 3
Grade 5
Grade 4
(x, y) = (0.683, −0.721)


Figure 1
Row vs Enrollment
800000
600000
400000
200000
0
Enrollment
0
10
20
30
40
50
Row
(x, y) = (29.7, 3.37e+05)


Figure 1
Enrollment Distribution
30
25
20
15
10
5
0
Frequency
0
200000
400000
600000
800000
Enrollment
(x, y) = (7.39e+05, 10.16)





5. I feel like the best chart would be a line chart to communicate the findings as this chart shows the yearly trends of the enrolments.

6. Provides more control and comes with more code to write compared to other library

7. Matplotlib should add more designing for much more flexible visualization and easier data presentation

8. Matplotlib gives basic plotting functions like line, bar, and pie chart which are intuitive that only need a line or two of code to produce a working chart.

```python
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('combined_enrollment.csv')

dataset = data.iloc[:50, :7]
print(dataset)

rows = dataset.index

plt.plot(rows, dataset['Enrollment'], marker='o')
plt.title("Enrollment per row")
plt.xlabel("Row")
plt.ylabel("Enrollment")
plt.show()

plt.bar(rows, dataset['Enrollment'])
plt.title("Enrollment per row")
plt.xlabel("Row")
plt.ylabel("Enrollment")
plt.show()

plt.barh(dataset.index[:15].astype(str), dataset['Enrollment'].iloc[:15])
plt.title("Horizontal bar")
plt.show()

genderCount = dataset['Gender'].value_counts()
plt.pie(genderCount, labels=genderCount.index.astype(str))
plt.title("Gender Distribution")
plt.show()

gradeCount = dataset['Grade/Strand'].value_counts()
explode = [0.1] * len(gradeCount)
plt.pie(gradeCount, labels=gradeCount.index.astype(str), explode=explode)
plt.title("Grade/Strand distribution")
plt.show()

plt.scatter(rows, dataset['Enrollment'])
plt.title("Row vs Enrollment")
plt.xlabel("Row")
plt.ylabel("Enrollment")
plt.show()

plt.hist(dataset['Enrollment'], bins=10)
plt.title("Enrollment Distribution")
plt.xlabel("Enrollment")
plt.ylabel("Frequency")
plt.show()
```

```
49
50  plt.fill_between(range(len(dataset)), dataset['Enrollment'], alpha=0.6)
51  plt.plot(range(len(dataset)), dataset['Enrollment'])
52  plt.title("Enrollment area chart")
53  plt.xlabel("Row")
54  plt.ylabel("Enrollment")
55  plt.show()
56
57  # split the existing Gender column into 3 series for the stacked area
58  # (these are just temporary variables, not new columns in dataset)
59  male = (dataset['Gender'] == 'Male').astype(int)
60  female = (dataset['Gender'] == 'Female').astype(int)
61  total = (dataset['Gender'] == 'Total').astype(int)
62
63  plt.stackplot(range(len(dataset)), male, female, total,
64                labels=['Male', 'Female', 'Total'])
65  plt.title("Stacked Area: Gender per row")
66  plt.xlabel("Row")
67  plt.ylabel("Value")
68  plt.legend(loc='upper left')
69  plt.show()
```